Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 5

Converse, Contrapositive, and the Language of "If"

Covers Chartrand & Zhang, Section 1.3, pp. 28–32

Status of this set

Ungraded practice. Solutions released with it. Attempt everything before reading any solution.

Part I: How to Practice This Material

Two questions to ask every time

The drill

When you meet an implication, ask:

1. **Which way does the arrow point?** English hides this. "P only if Q" and "Q if P" both mean $P \Rightarrow Q$, even though Q is written first in one of them.
2. **Am I being asked about the statement or its converse?** They are different claims, and confusing them is the most consequential error in Chapter 1.

The pairing, in one line

$$P \Rightarrow Q \;\equiv\; (\sim Q) \Rightarrow (\sim P) \qquad \text{and} \qquad Q \Rightarrow P \;\equiv\; (\sim P) \Rightarrow (\sim Q)$$

Implication pairs with contrapositive; converse pairs with inverse. A single counterexample kills both members of a pair at once.

Three mistakes to expect

1. **Reading "P only if Q" as $Q \Rightarrow P$.** It is $P \Rightarrow Q$.
2. **Swapping "necessary" and "sufficient."** In $P \Rightarrow Q$, the hypothesis is sufficient and the conclusion is necessary.
3. **Forming the contrapositive by negating only.** That gives the inverse. You must negate *and* swap.

Part II: Practice Problems

Section A — Reading the arrow

1. Rewrite each as an "if. . . then" sentence.

(a) Having a valid ticket is necessary to board the train.

(b) Finishing the project early is sufficient to earn the bonus.

(c) The alarm sounds only if the door is opened.

(d) I will review your draft if you send it by Thursday.

2. Express the implication "If $ab = 0$, then $a = 0$ or $b = 0$" in each of these ways: (a) using "only if"; (b) using "sufficient"; (c) using "necessary."

Section B — The four forms

3. For the implication *"If a quadrilateral is a square, then it is a rectangle,"* write the converse, the inverse, and the contrapositive, and state whether each is true.

4. For an integer n, consider: *"If n is divisible by* 10*, then n is even."*

(a) Write the contrapositive.

(b) Write the converse.

(c) One of (a) and (b) is true and the other is false. Say which, and give a counterexample for the false one.

Section C — Where it bites

5. A diagnostic test satisfies: *"If a patient has the disease, then the test is positive."*

(a) Write the contrapositive, and say in plain words what it tells a patient.

(b) Write the converse.

(c) A patient tests positive. Does the stated property let you conclude the patient has the disease? Explain.

6. A codebase has the documented invariant: *"If a connection is open, then it appears in the active-connections table."*

(a) A connection does not appear in the table. What may you conclude?

(b) A connection appears in the table. What may you conclude?

Section D — Writing

7. In five to eight complete sentences, explain why proving the contrapositive of a statement counts as proving the statement itself, while proving the converse does not. Use the truth table from class, and give one concrete pair of statements to illustrate.

Section E — Challenge (optional)

8. Show that the converse and the inverse of $P \Rightarrow Q$ are logically equivalent to each other, *without* building a truth table. Use Theorem 1.46.